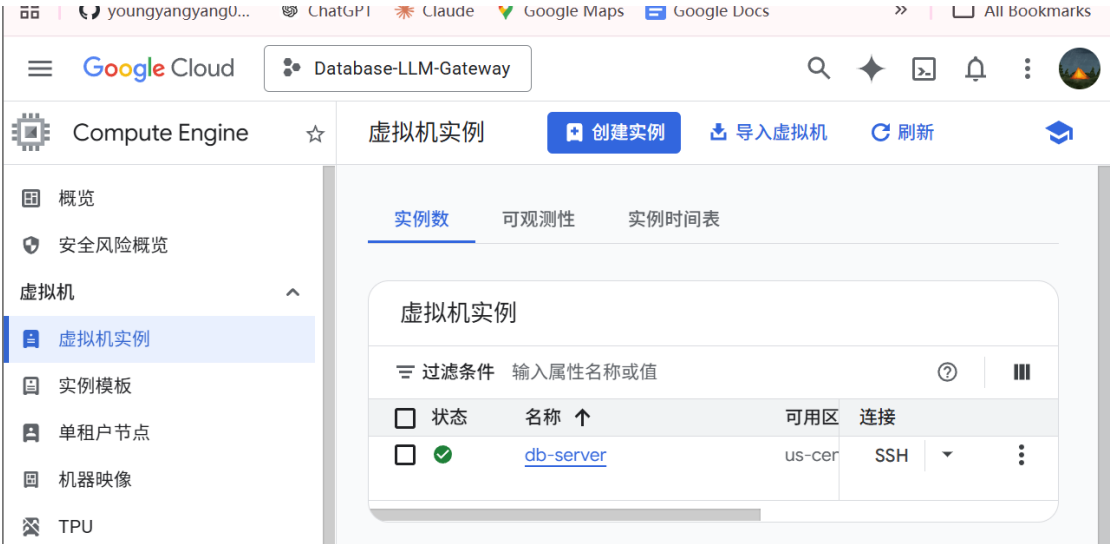


Group Phase 3 - System Setup using GCP

Step 8: Creating Database Server (PostgreSQL on Compute Engine)

8.1 GCP Project Creation



A new GCP project named “Database-LLM-Gateway” was created via the GCP Console.

A Compute Engine VM instance named “db-server” was created using Ubuntu 20.04 LTS with machine type e2-medium. This VM serves as the dedicated database server for the project.

8.3 PostgreSQL Installation

PostgreSQL was installed on the Compute Engine VM via SSH. The following commands were used to install and start the database service:

```
sudo apt update && sudo apt install -y postgresql  
  
sudo systemctl start postgresql && sudo systemctl enable postgresql
```

The screenshot below confirms PostgreSQL is running successfully on the VM (status: active).

SSH-in-browser [↑](#) UPLOAD FILE [↓](#) DOWNLOAD FILE [?](#) [⌨](#) [⚙](#)

```
Creating config file /etc/postgresql-common/createcluster.conf with new version
Building PostgreSQL dictionaries from installed myspell/hunspell packages...
Removing obsolete dictionary files:
Created symlink /etc/systemd/system/multi-user.target.wants/postgresql.service → /lib/systemd/system/postgresql.service.
Setting up libtypes-serialiser-perl (1.01-1) ...
Setting up postgresql-15 (15.16-0+deb12u1) ...
Creating new PostgreSQL cluster 15/main ...
/usr/lib/postgresql/15/bin/initdb -D /var/lib/postgresql/15/main --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are disabled.

fixing permissions on existing directory /var/lib/postgresql/15/main ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default max connections ... 100
selecting default shared buffers ... 128MB
selecting default time zone ... Etc/UTC
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
update-alternatives: using /usr/share/postgresql/15/man/man1/postmaster.1.gz to provide /usr/share/man/man1/postmaster.1.gz (postmaster.1.gz) in auto mode
Setting up libjson-xs-perl (4.040-1~deb12u1) ...
Setting up postgresql-contrib (15+248+deb12u1) ...
Setting up postgresql (15+248+deb12u1) ...
Processing triggers for man-db (2.11.2-2) ...
Processing triggers for libc-bin (2.36-9+deb12u13) ...
Synchronizing state of postgresql.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable postgresql
shaorunxinb@db-server:~$
```

Figure 1: PostgreSQL service status showing active (running) on the Compute Engine VM.

VPC firewall rules

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. [Learn more](#)

Note: App Engine firewalls are managed in the [App Engine Firewall rules section](#)

SMTP port 25 disallowed in this project. [Learn more](#)

[Refresh](#) [Configure logs](#) [Delete](#)

Filter Enter property name or value [?](#) [⌵](#)

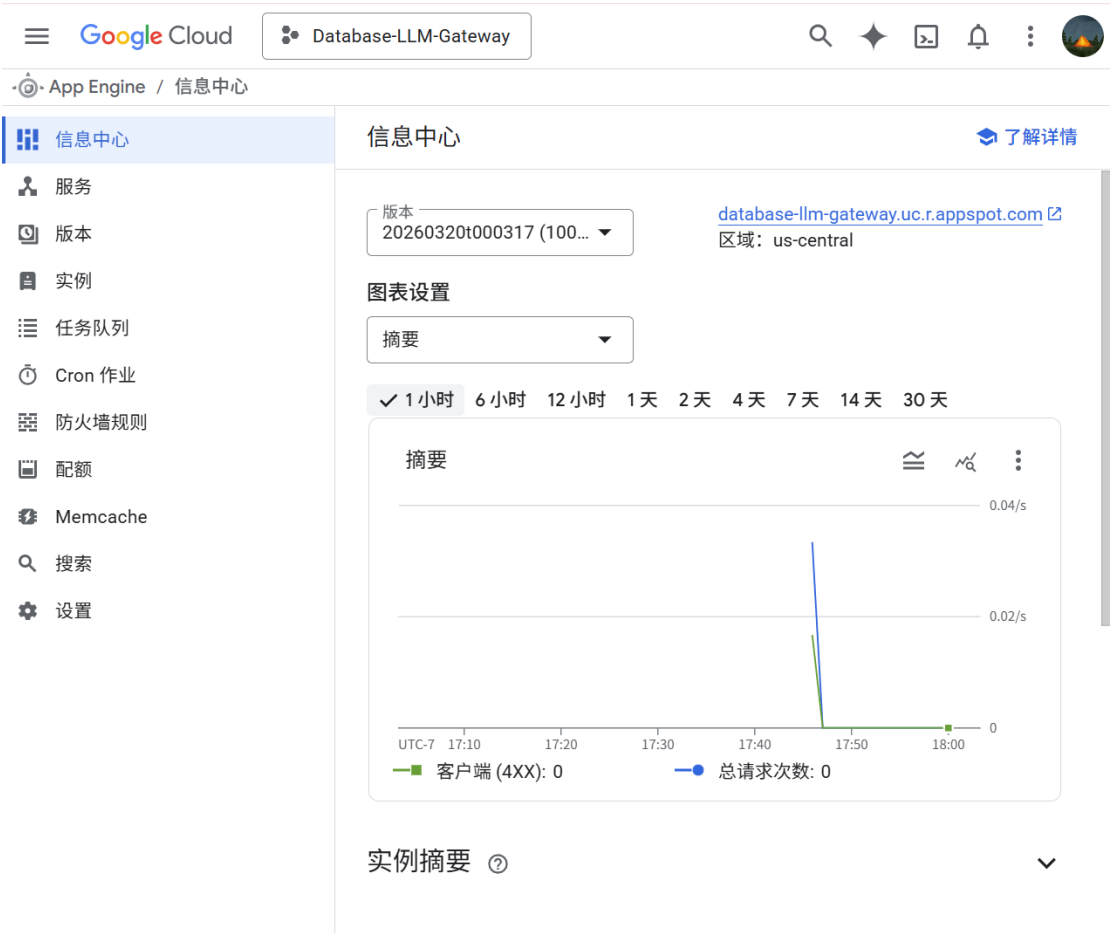
☐	Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network ↑	Logs	⌵
☐	allow-db-port	Ingress	Apply to all	IP ranges:	tcp:5432	Allow	1000	default	Off	⌵
☐	default-allow-http	Ingress	http-server	IP ranges:	tcp:80	Allow	1000	default	Off	⌵
☐	default-allow-https	Ingress	https-server	IP ranges:	tcp:443	Allow	1000	default	Off	⌵
☐	default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	65534	default	Off	⌵
☐	default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	65534	default	Off	⌵
☐	default-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	65534	default	Off	⌵
☐	default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	65534	default	Off	⌵

Figure 2: Firewall rule allowing TCP port 5432 for PostgreSQL inbound traffic.

Figure 3: Firewall rule details confirming port 5432 is open.

Step 9: Creating Web Application Server (Spring Boot on App Engine)

9.1 App Engine Setup



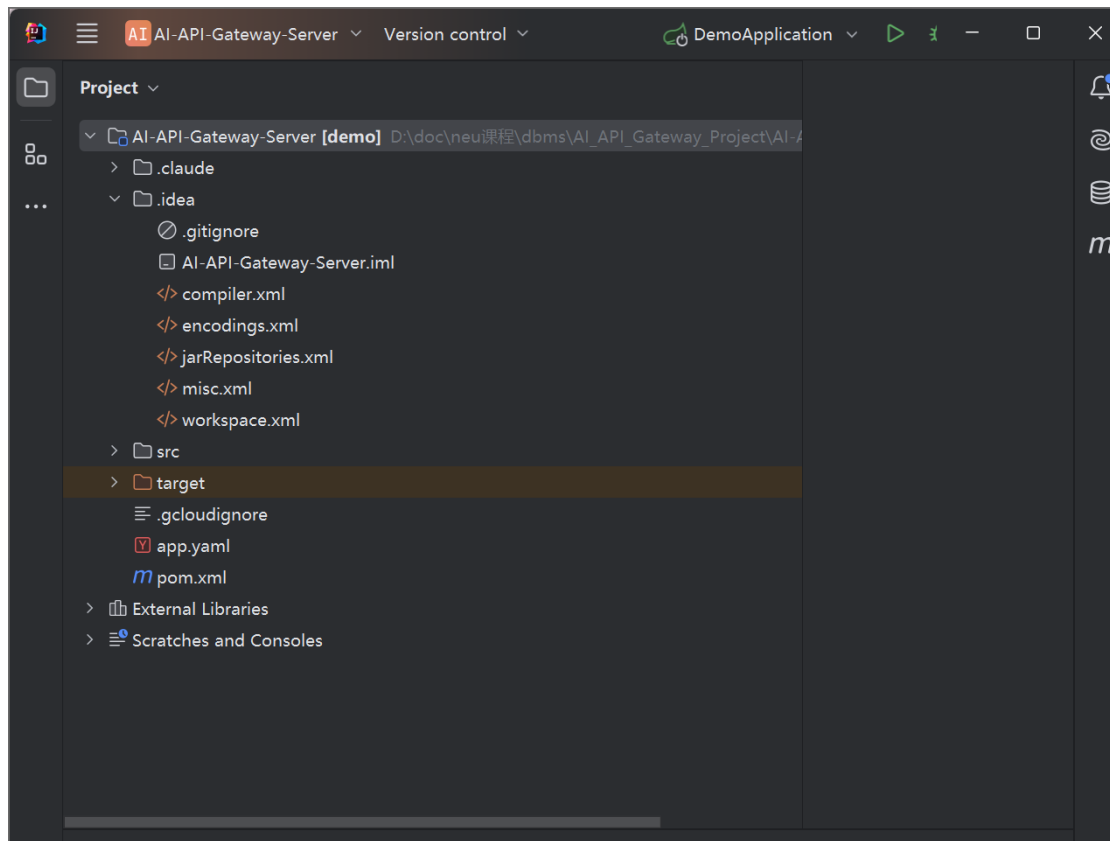
Google App Engine was enabled for the project. A Java 17 standard environment application was created in the us-central region to host the Spring Boot web application.

9.2 Spring Boot Application Deployment

A Spring Boot application was built using Spring Initializr with Spring Web, Spring Data JPA, and PostgreSQL Driver dependencies. An app.yaml configuration file was created for App Engine deployment. The application was packaged and deployed using the following commands:

```
mvn clean package -DskipTests
```

```
mvn appengine:deploy
```



9.3 App Engine Services and Deployed Version



9.4 Web Application Running in Browser

After deployment, the application was accessed via the App Engine URL.

<https://database-llm-gateway.uc.r.appspot.com/>

The screenshot below confirms the web application is live and accessible.

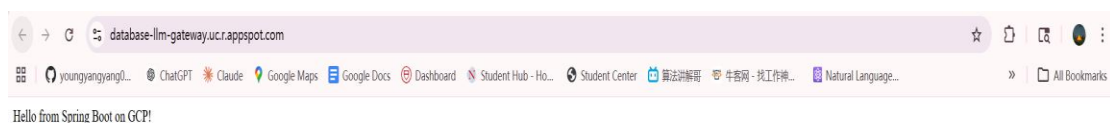


Figure 4: Spring Boot web application running and accessible on App Engine URL.

